

dotnet add package Npgsql

Appsettings.json

```
1  {
2    "ConnectionStrings": {
3      "pgconn": "Server=localhost;Port=5432;Database=stud2025;User Id=postgres;Password=Admin@123;"
4    },
5    "Logging": {
6      "LogLevel": {
7        "Default": "Information",
8        "Microsoft.AspNetCore": "Warning"
9      }
10   },
11   "AllowedHosts": "*"
12 }
13
```

Program.cs

```
...
builder.Services.AddSingleton<NpgsqlConnection>((conn) =>
{
    var connectionString = conn.GetRequiredService<IConfiguration>().GetConnectionString("pgconn");
    return new NpgsqlConnection(connectionString);
});
builder.Services.AddSession(options =>
{
    options.IdleTimeout = TimeSpan.FromMinutes(30);
    options.Cookie.HttpOnly = true;
    options.Cookie.IsEssential = true;
});
...
```

Models/Student.cs

```
7  using System.ComponentModel.DataAnnotations;
8
9  namespace MVCApp.Models
10 {
11     17 references
12     public class Student
13     {
14         9 references
15         public int c_studid { get; set; }
16
17         [Required(ErrorMessage = "Student name is required.")]
18         [RegularExpression(@"^[a-zA-Z0-9 ]+$", ErrorMessage = "Only alphanumeric characters and spaces are allowed.")]
19         12 references
20         public string c_studname { get; set; }
21
22         [DataType(DataType.Date)]
23         12 references
24         public DateTime? c_dateofbirth { get; set; }
25
26         10 references
27         public string c_stream { get; set; }
28     }
29 }
```

Services/StudentService.cs

```
1 using Npgsql;
2 using MVCApp.Models;
3
4 namespace MVCApp.Services
5 {
6     4 references
7     public class StudentService
8     {
9         16 references
10        private readonly NpgsqlConnection conn;
11
12        0 references
13        public StudentService(NpgsqlConnection connection)
14        {
15            conn = connection;
16        }
17
18        1 reference
19        public async Task<List<Student>> GetAllAsync()
20        {
21            var students = new List<Student>();
22            try
23            {
24                await conn.OpenAsync();
25
26                using var cmd = new NpgsqlCommand("SELECT c_studid, c_studname, c_dateofbirth, c_stream FROM t_student ORDER BY c_studid", conn);
27                using var reader = await cmd.ExecuteReaderAsync();
28
29                while (await reader.ReadAsync())
30                {
31                    students.Add(new Student
32                    {
33                        c_studid = reader.GetInt32(0),
34                        c_studname = reader.GetString(1),
35                        c_dateofbirth = reader.IsDBNull(2) ? null : reader.GetDateTime(2),
36                        c_stream = reader.IsDBNull(3) ? null : reader.GetString(3)
37                    });
38                }
39            }
40            catch (Exception ex) {}
41            finally
42            {
43                await conn.CloseAsync();
44            }
45            return students;
46        }
47    }
48 }
```

```

2 references
44 public async Task<Student> GetByIdAsync(int id)
45 {
46     try
47     {
48         await conn.OpenAsync();
49         using var cmd = new NpgsqlCommand("SELECT c_studid, c_studname, c_dateofbirth, c_stream FROM t_student WHERE c_studid = @id", conn);
50         cmd.Parameters.AddWithValue("id", id);
51         using var reader = await cmd.ExecuteReaderAsync();
52         if (await reader.ReadAsync())
53         {
54             return new Student
55             {
56                 c_studid = reader.GetInt32(0),
57                 c_studname = reader.GetString(1),
58                 c_dateofbirth = reader.IsDBNull(2) ? null : reader.GetDateTime(2),
59                 c_stream = reader.IsDBNull(3) ? null : reader.GetString(3)
60             };
61         }
62     }
63     catch (Exception ex) {}
64     finally
65     {
66         await conn.CloseAsync();
67     }
68     return null;
69 }
70

```

```

71 public async Task AddAsync(Student student)
72 {
73     try
74     {
75         await conn.OpenAsync();
76
77         using var cmd = new NpgsqlCommand(@"INSERT INTO t_student (c_studname, c_dateofbirth, c_Stream)
78         VALUES (@name, @dob, @stream)", conn);
79         cmd.Parameters.AddWithValue("name", student.c_studname);
80         cmd.Parameters.AddWithValue("dob", (object?)student.c_dateofbirth ?? DBNull.Value);
81         cmd.Parameters.AddWithValue("stream", (object?)student.c_stream ?? DBNull.Value);
82
83         await cmd.ExecuteNonQueryAsync();
84     }
85     catch (Exception ex) { }
86     finally
87     {
88         await conn.CloseAsync();
89     }
90 }
91
92

```

```

1 reference
93 public async Task UpdateAsync(Student student)
94 {
95     try
96     {
97         await conn.OpenAsync();
98
99         using var cmd = new NpgsqlCommand(@"UPDATE t_student SET c_studname = @name, c_dateofbirth = @dob, c_stream = @stream
100 WHERE c_studid = @id", conn);
101         cmd.Parameters.AddWithValue("name", student.c_studname);
102         cmd.Parameters.AddWithValue("dob", (object?)student.c_dateofbirth ?? DBNull.Value);
103         cmd.Parameters.AddWithValue("stream", (object?)student.c_stream ?? DBNull.Value);
104         cmd.Parameters.AddWithValue("id", student.c_studid);
105
106         await cmd.ExecuteNonQueryAsync();
107     }
108     catch (Exception ex) { }
109     finally
110     {
111         await conn.CloseAsync();
112     }
113 }
114

```

```
1 reference
115 public async Task DeleteAsync(int id)
116 {
117     try
118     {
119         await conn.OpenAsync();
120
121         using var cmd = new NpgsqlCommand("DELETE FROM t_student WHERE c_studid = @id", conn);
122         cmd.Parameters.AddWithValue("id", id);
123
124         await cmd.ExecuteNonQueryAsync();
125     }
126     catch (Exception ex) { }
127     finally
128     {
129         await conn.CloseAsync();
130     }
131 }
132 }
133 }
134
```

Program.cs

```
...
builder.Services.AddSingleton<StudentService>();
...
```

StudentController.cs

```
1 |
2 | using Microsoft.AspNetCore.Mvc;
3 | using MVCApp.Models;
4 | using MVCApp.Services;
5 | namespace MVCApp.Controllers
6 | {
7 |     1 reference
8 |     public class StudentController : Controller
9 |     {
10 |         7 references
11 |         private readonly StudentService _studentService;
12 |         0 references
13 |         public StudentController(StudentService studentService)
14 |         {
15 |             _studentService = studentService;
16 |         }
17 |         0 references
18 |         public async Task<IActionResult> Index()
19 |         {
20 |             var students = await _studentService.GetAllAsync();
21 |             return View(students);
22 |         }
23 |         0 references
24 |         public IActionResult Create()
25 |         {
26 |             return View();
27 |         }
28 |         [HttpPost]
29 |         0 references
30 |         public async Task<IActionResult> Create(Student student)
31 |         {
32 |             if (ModelState.IsValid)
33 |             {
34 |                 await _studentService.AddAsync(student);
35 |                 return RedirectToAction("Index");
36 |             }
37 |             return View(student);
38 |         }
39 |         0 references
40 |         public async Task<IActionResult> Edit(int id)
41 |         {
42 |             var student = await _studentService.GetByIdAsync(id);
43 |             if (student == null)
44 |                 return NotFound();
45 |             return View(student);
46 |         }
47 |     }
48 | }
```

```

42 | 0 references
43 | public async Task<IActionResult> Edit(int id)
44 | {
45 |     var student = await _studentService.GetByIdAsync(id);
46 |     if (student == null)
47 |         return NotFound();
48 |     return View(student);
49 | }
50 |
51 | [HttpPost]
52 | 0 references
53 | public async Task<IActionResult> Edit(Student student)
54 | {
55 |     if (ModelState.IsValid)
56 |     {
57 |         await _studentService.UpdateAsync(student);
58 |         return RedirectToAction("Index");
59 |     }
60 |     return View(student);
61 | }
62 | 0 references
63 | public async Task<IActionResult> Delete(int id)
64 | {
65 |     var student = await _studentService.GetByIdAsync(id);
66 |     if (student == null)
67 |         return NotFound();
68 |     return View(student);
69 | }
70 |
71 | [HttpPost, ActionName("Delete")]
72 | 0 references
73 | public async Task<IActionResult> DeleteConfirmed(int id)
74 | {
75 |     await _studentService.DeleteAsync(id);
76 |     return RedirectToAction("Index");
77 | }
78 | }

```

_StudentLayout.cshtml

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4     <meta charset="utf-8" />
5     <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6     <title>@ViewData[index: "Title"] - Student</title>
7     <link rel="stylesheet" href="~/lib/bootstrap/dist/css/bootstrap.min.css" />
8     <link rel="stylesheet" href="~/css/site.css" asp-append-version="true" />
9     <link rel="stylesheet" href="~/MVCAApp.styles.css" asp-append-version="true" />
10 </head>
11 <body>
12     <header>
13         <nav class="navbar navbar-expand-sm navbar-toggleable-sm navbar-light bg-white border-bottom box-shadow mb-3">
14             <div class="container-fluid">
15                 <a class="navbar-brand" asp-area="" asp-controller="Home" asp-action="Index">Student</a>
16                 <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target=".navbar-collapse" aria-controls="navbarSupportedContent"
17                     aria-expanded="false" aria-label="Toggle navigation">
18                     <span class="navbar-toggler-icon"></span>
19                 </button>
20                 <div class="navbar-collapse collapse d-sm-inline-flex justify-content-between">
21                     <ul class="navbar-nav flex-grow-1">
22                         <li class="nav-item">
23                             <a class="nav-link text-dark" asp-area="" asp-controller="Student" asp-action="Index">List</a>
24                         </li>
25                         <li class="nav-item">
26                             <a class="nav-link text-dark" asp-area="" asp-controller="Student" asp-action="Create">New</a>
27                         </li>
28                     </ul>
29                 </div>
30             </div>
31         </nav>
32     </header>
33     <div class="container">
34         <main role="main" class="pb-3">
35             @RenderBody()
36         </main>
37     </div>
38     <footer class="border-top footer text-muted">
39         <div class="container">
40             &copy; 2025 - MVCAApp - <a asp-area="" asp-controller="Home" asp-action="Privacy">Privacy</a>
41         </div>
42     </footer>
43     <script src="~/lib/jquery/dist/jquery.min.js"></script>
44     <script src="~/lib/bootstrap/dist/js/bootstrap.bundle.min.js"></script>
45     <script src="~/js/site.js" asp-append-version="true"></script>
46     @await RenderSectionAsync(name: "Scripts", required: false)
47 </body>
48 </html>
```

Index.cshtml

```
1 @model List<MVCAApp.Models.Student>
2 @{
3     ViewData[index: "Title"] = "List";
4     Layout = "~/Views/Shared/StudentLayout.cshtml";
5 }
6 <h2>Students</h2>
7 <p>
8     <a asp-action="Create" class="btn btn-primary">Add New Student</a>
9 </p>
10 <table class="table table-bordered">
11     <thead>
12         <tr>
13             <th>ID</th>
14             <th>Name</th>
15             <th>Date of Birth</th>
16             <th>Stream</th>
17             <th>Actions</th>
18         </tr>
19     </thead>
20     <tbody>
21         @foreach (var Student student in Model)
22         {
23             <tr>
24                 <td>@student.c_studid</td>
25                 <td>@student.c_studname</td>
26                 <td>@student.c_dateofbirth?.ToString(format: "yyyy-MM-dd")</td>
27                 <td>@student.c_stream</td>
28                 <td>
29                     <a asp-action="Edit" asp-route-id="@student.c_studid" class="btn btn-sm btn-primary">Edit</a>
30                     <a asp-action="Delete" asp-route-id="@student.c_studid" class="btn btn-sm btn-danger">Delete</a>
31                 </td>
32             </tr>
33         }
34     </tbody>
35 </table>
```

Create.cshtml

```
1 @model MVCApp.Models.Student
2 @{
3     ViewData[index: "Title"] = "Create";
4     Layout = "~/Views/Shared/StudentLayout.cshtml";
5 }
6 <h2>Create Student</h2>
7 <form asp-action="Create" method="post">
8     <div class="form-group">
9         <label asp-for="c_studname">Student Name</label>
10        <input asp-for="c_studname" class="form-control" />
11        <span asp-validation-for="c_studname" class="text-danger"></span>
12    </div>
13
14    <div class="form-group">
15        <label asp-for="c_dateofbirth">Date of Birth</label>
16        <input asp-for="c_dateofbirth" type="date" class="form-control" />
17        <span asp-validation-for="c_dateofbirth" class="text-danger"></span>
18    </div>
19
20    <div class="form-group">
21        <label asp-for="c_stream">Stream</label>
22        <input asp-for="c_stream" class="form-control" />
23    </div>
24
25    <button type="submit" class="btn btn-primary">Create</button>
26 </form>
27
28 @section Scripts {
29     @{await Html.RenderPartialAsync(partialViewName: "_ValidationScriptsPartial");}
30 }
31
```

Edit.cshtml

```
1 @model MVCApp.Models.Student
2 @{
3     ViewData[index: "Title"] = "Edit";
4     Layout = "~/Views/Shared/StudentLayout.cshtml";
5 }
6 <h2>Edit Student</h2>
7 <form asp-action="Edit" method="post">
8     <input type="hidden" asp-for="c_studid" />
9
10    <div class="form-group">
11        <label asp-for="c_studname">Student Name</label>
12        <input asp-for="c_studname" class="form-control" />
13        <span asp-validation-for="c_studname" class="text-danger"></span>
14    </div>
15
16    <div class="form-group">
17        <label asp-for="c_dateofbirth">Date of Birth</label>
18        <input asp-for="c_dateofbirth" type="date" class="form-control" />
19        <span asp-validation-for="c_dateofbirth" class="text-danger"></span>
20    </div>
21
22    <div class="form-group">
23        <label asp-for="c_stream">Stream</label>
24        <input asp-for="c_stream" class="form-control" />
25    </div>
26
27    <button type="submit" class="btn btn-primary">Update</button>
28    <a asp-action="Index" class="btn btn-secondary">Back</a>
29 </form>
30
31 @section Scripts {
32     @{await Html.RenderPartialAsync(partialViewName: "_ValidationScriptsPartial");}
33 }
34
```

Delete.cshtml

```
1  @model MVCApp.Models.Student
2  @{
3      ViewData[index: "Title"] = "Delete";
4      Layout = "~/Views/Shared/StudentLayout.cshtml";
5  }
6  <h2>Delete Student</h2>
7  <h3>Are you sure you want to delete this student?</h3>
8  <div>
9      <h4>Student Details</h4>
10     <dl class="row">
11         <dt class="col-sm-2">ID</dt>
12         <dd class="col-sm-10">@Model.c_studid</dd>
13
14         <dt class="col-sm-2">Name</dt>
15         <dd class="col-sm-10">@Model.c_studname</dd>
16
17         <dt class="col-sm-2">Date of Birth</dt>
18         <dd class="col-sm-10">@Model.c_dateofbirth?.ToString(format: "yyyy-MM-dd")</dd>
19
20         <dt class="col-sm-2">Stream</dt>
21         <dd class="col-sm-10">@Model.c_stream</dd>
22     </dl>
23
24     <form asp-action="Delete" method="post">
25         <input type="hidden" asp-for="c_studid" />
26         <button type="submit" class="btn btn-danger">Delete</button>
27         <a asp-action="Index" class="btn btn-secondary">Cancel</a>
28     </form>
29 </div>
30
```